01 / PROBLEM AND CONTRIBUTION

Emergency-vehicle traffic signal prototype

Measurements & Instrumentation mini-project | Undergraduate | Three-person team

RESULT AT A GLANCE

Demonstrated IR-triggered signal selection on an Arduino/LabVIEW prototype: the selected direction turns green while the other directions turn red.

The engineering question

How can a detected vehicle request change a small traffic-light prototype's signal state?

Situation

The prototype explores how a vehicle request can change a traffic signal. An IR emitter represents the approaching emergency vehicle, with receivers connected to the signal-control system.

Task

Implement the Arduino control and contribute the report and presentation, working with the team's LabVIEW and hardware integration.

My contribution

Arduino implementation, report, and presentation; the team shared LabVIEW and hardware integration.

Tools and concepts

Arduino Uno / NI LabVIEW / IR sensing / Embedded control / Prototyping

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation step

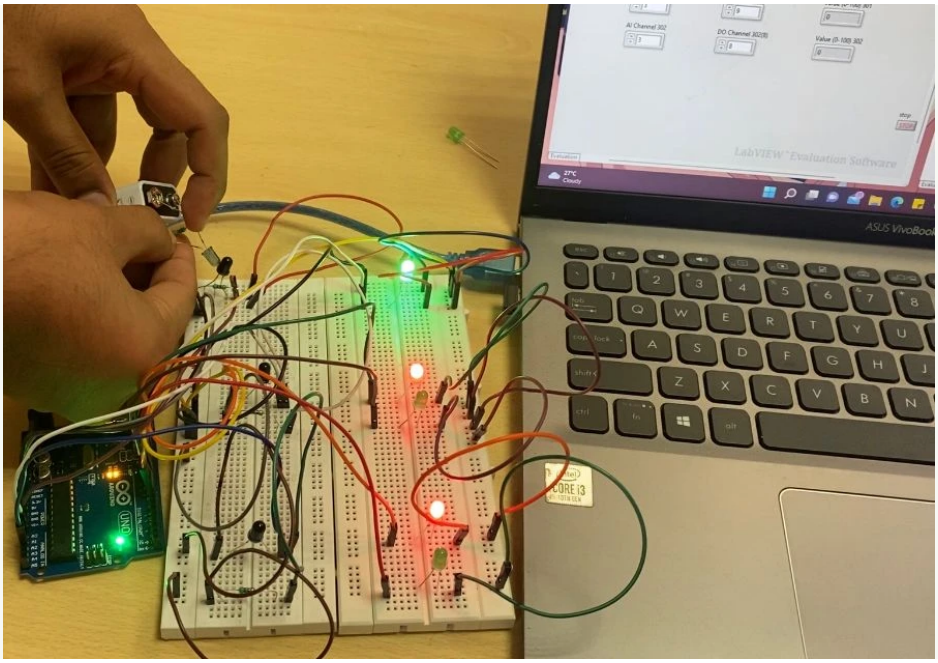
Used an IR emitter representing the approaching emergency vehicle and receivers at the signal. Arduino and LabVIEW logic translate detection into the light outputs.

02 / Implementation step

The selected direction turns green while the other directions turn red. The report includes the circuit, flowchart, result images, and component list.

Detection to output selection

An IR request is translated through Arduino/LabVIEW control into the selected green indication and red indications for the other directions. The circuit and result images document this prototype sequence.



Traffic-priority prototype

Arduino and breadboard demonstration photographed in the team report.

03 / RESULTS AND ENGINEERING PERSPECTIVE

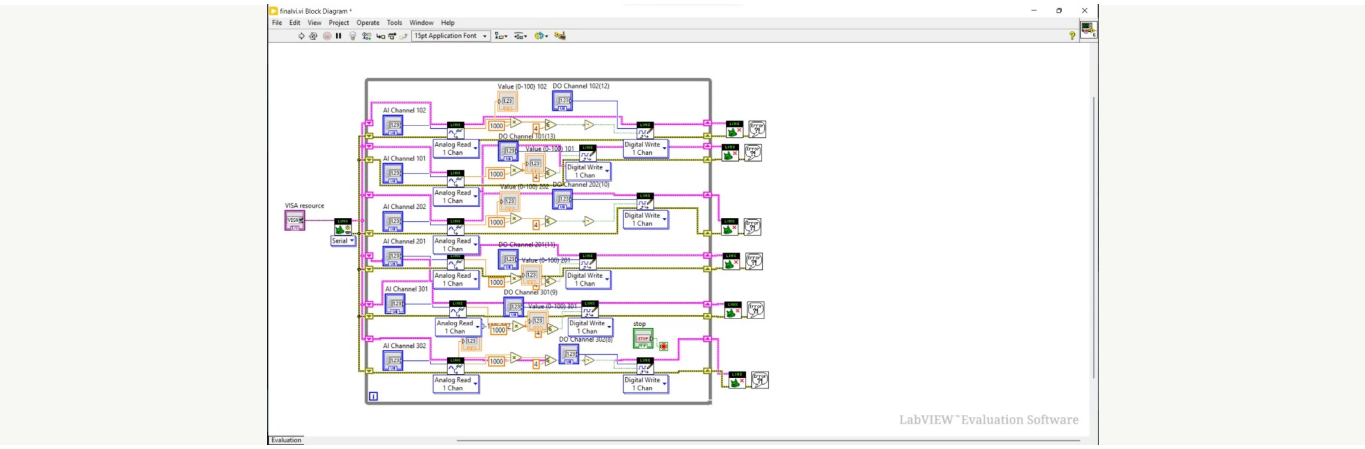
Evidence, interpretation, and next steps

Reported result

The project demonstrates the signal-selection behavior in a prototype. The contribution table identifies Ruthvik's Arduino and documentation work.

Prototype boundary

The demonstration concerns signal selection under a request. Multiple simultaneous emergency requests need arbitration, and the supplied design does not define that priority policy. No real-road outcome follows from the prototype alone.



LabVIEW implementation

LabVIEW program view from the project documentation.

Scope and limitations

The reported design does not assign priority when emergency vehicles arrive from multiple directions. No real-road deployment or measured response-time reduction is claimed.

Possible next steps - proposed, not completed results

• Define and test an arbitration policy for competing simultaneous requests.

• Measure request-to-output latency in the prototype before claiming a response-time improvement.

Source material: Emergency vehicle traffic clearing system report in gptz.zip, working principle, results, limitations, and contributions.

Prepared from the supplied reports, project files, and verified portfolio copy. Figures retain their original content; explanatory diagrams are labelled.

Project files and documentation: Emergency-vehicle-traffic-clearing-system-using-LABVIEW-and-Arduino